Post-selection inference in multiverse meta-analysis

Matteo Manente[1], Livio Finos[2], Gianmarco Altoè[1], and Filippo Gambarota[1]

[1]Department of Developmental Psychology and Socialization, University of Padova

[2]Department of Statistical Sciences, University of Padova

Author Note

Correspondence concerning this article should be addressed to Filippo Gambarota, Department of Developmental Psychology and Socialization, University of Padova, Via Venezia 8, Padova, Italy 35131, Email: filippo.gambarota@unipd.it

Abstract

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Introduction

Doing research can be hard. Researchers must make many decisions during a study. These decisions concern choosing between different measurement instruments, data pre-processing procedures, and statistical models. Each decision is usually reasonable and defensible but often arbitrary. Together, these decisions are known as *researcher degrees of freedom* (Simmons et al., 2011).

Researcher degrees of freedom are especially common in psychology and the social sciences, but they also occur in other fields such as medicine and biology (Errington et al., 2021; Nosek et al., 2022). In psychology, for example, phenomena are often difficult to define and causal relationships between variables are difficult to establish (Eronen & Bringmann, 2021). There are numerous measurement instruments, which are often poorly validated (Flake & Fried, 2020), while data analysis is complex and rarely guided by a single optimal solution. Together, these features make it difficult to justify one analytic approach over another. In practice, there is often no single best procedure for data collection, pre-processing, or analysis. Researchers must therefore choose one path among many valid ones in the so-called "garden of forking paths" (Gelman & Loken, 2013). Importantly, different choices can lead to different results. Even with the same data and research question, different analytical strategies may yield different conclusions (Silberzahn et al., 2018). Researchers therefore often face a *multiverse* of equally reasonable outcomes (Steegen et al., 2016).

The existence of researcher degrees of freedom does not necessarily constitute a statistical problem when analytical choices are fully determined in advance. However, when multiple reasonable specifications are examined and the observed results influence which analyses are emphasized or interpreted, a multiple testing and selective-inference problem arises. This is particularly evident in questionable research practices such as *p-hacking*—testing many analytic options until a non-significant result becomes significant—but the same inferential issue can also arise without intentional manipulation when researchers explore multiple analyses and selectively interpret their results (Gelman & Loken, 2013). Without proper adjustment, this process can inflate the false positive rate beyond the nominal level (e.g., α = 5). Researcher degrees of freedom have therefore been considered one of the factors contributing to the replication crisis (Open Science Collaboration, 2015; Wicherts et al., 2016).

Importantly, researcher degrees of freedom are unavoidable. They are a natural part of empirical research. The problem lies not in their existence but in their opacity and in the selective interpretation of their results. When only one analytic path is reported, reviewers and readers cannot assess how results depend on alternative choices or how robust the findings truly are.

Instead of presenting a single path, *multiverse analysis* was proposed to make the full set of analytic choices and results transparent (Steegen et al., 2016). The core idea is simple: the data are processed and analyzed using all reasonable combinations of analytic choices, and the resulting estimates are jointly reported (Simonsohn et al., 2020; Steegen et al., 2016). This approach makes explicit the variability induced by analytic choices and allows readers to evaluate how conclusions depend on alternative decisions.

Although multiverse analysis was developed mainly for primary studies, meta-analyses face the same challenges. A *meta-analysis* aims to quantitatively synthesize evidence across studies (Borenstein et al., 2009). However, meta-analyses are also affected by researcher degrees of freedom. Researchers must choose, for example, which studies to include, which effect-size metrics to use, how to compute the sampling variance, and which statistical model to implement (e.g., fixed- versus random-effects models). Each decision can be reasonable, yet it can influence the final result. This becomes even more relevant in meta-analyses involving complex data structures, such as multilevel or multivariate data, which often require imputing missing information, aggregating effect sizes, or making additional modeling assumptions (Cheung, 2013; Van den Noortgate et al., 2013).

This issue is particularly important because systematic reviews and meta-analyses are traditionally placed at the top of the hierarchy of evidence and can substantially influence clinical and policy decisions (Higgins, 2019; Murad et al., 2016). Therefore, although assessing the robustness of results across plausible analytical choices is important in any empirical investigation, it is especially critical in meta-analysis. A dramatic change in conclusions following minor analytical changes suggests that the evidence is fragile, potentially due to weak theory, poor measurement, heterogeneous effects, or other factors.

Recently, several authors have assessed the robustness of research findings using multiverse meta-analysis, for example examining the efficacy of psychotherapy and digital interventions for depression (Plessen et al., 2023, 2025), the association between IQ and mental illness (Fries et al., 2025), the effects of oxytocin administration (Kang et al., 2025), and the effects of music listening on epilepsy [i.e., the Mozart effect; Oberleiter and Pietschnig (2023)].

However, despite these advantages, current approaches to multiverse analysis are still limited from an inferential perspective. Many are primarily descriptive, such as the original multiverse analysis proposed by (Steegen et al., 2016) or the *Vibration of Effects* framework (Patel et al., 2015). These methods visualize estimated parameters and p-values across the multiverse and provide summary measures of their variability. Although useful for assessing robustness, they do not provide a formal inferential answer to the original hypothesis-testing question. A research question initially formulated in confirmatory terms is therefore effectively shifted toward an exploratory description of the multiverse.

In inferential terms, a multiverse can be viewed as a multiple testing problem (Girardi et al., 2024; Goeman & Solari, 2014; Westfall & Young, 1992). Multiple statistical tests are performed on the same data, and the resulting test statistics are generally strongly dependent because specifications often differ only by a small number of analytical choices. Interpreting unadjusted p-values across such specifications can therefore be misleading, even when they are initially presented for exploratory purposes.

Inference across multiple specifications can be conducted either globally, by testing an omnibus hypothesis about the complete set of results, or selectively, by making inference on particular specifications. A global test considers the intersection null hypothesis that the effect is null in all specifications. Rejecting this hypothesis with a given error rate (e.g., 5%) allows researchers to conclude that the null hypothesis is false in at least one specification, but does not identify which specification is responsible for the rejection. With hundreds or thousands of specifications, this may provide only limited substantive insight (Oberauer & Lewandowsky, 2019, p. 1609). By contrast, making inferential claims about one or more specifications selected after inspecting the multiverse requires multiplicity-adjusted or post-selection inference, introducing an important trade-off between false positive error control and statistical power.

Currently available inferential methods for multiverse analysis are limited and have mostly focused on global inference. For example, *Specification Curve Analysis* (SCA) (Simonsohn et al., 2020) is one of the most widely used inferential multiverse tools and has also been applied in multiverse meta-analysis (see, e.g., Pietschnig et al., 2022; Plessen et al., 2023). SCA uses resampling under the null to generate a joint reference distribution of results across a set of reasonable analytical specifications, allowing the observed specification curve to be compared with what would be expected if the effect were absent. However, its inferential procedure is primarily developed for linear-model settings and does not provide a general framework for selective inference on individual specifications.

More recently, the *Post-selection Inference in Multiverse Analysis* (PIMA) framework (Girardi et al., 2024) provided a general inferential approach for multiverse analysis. PIMA is based on sign-flipping score tests and combines information across specifications through a joint resampling procedure. It allows researchers to test a global null hypothesis across the multiverse and, importantly, to perform selective inference on individual specifications while strongly controlling the family-wise error rate. Compared with SCA, PIMA was developed within the generalized linear model framework and therefore applies to a wider range of outcome distributions and model specifications.

Extending this framework to meta-analysis, however, is not straightforward. Multiverse meta-analysis combines two distinct inferential problems. First, inference *within* each specification may itself be poorly calibrated. Standard Wald-type inference in random- and mixed-effects meta-analytic models relies on large-sample approximations with respect to the number of studies, and these approximations can perform poorly when the number of studies is small or moderate (Follmann & Proschan, 1999; Viechtbauer et al., 2015). Second, inference *across* specifications introduces a multiple testing and post-selection problem because a large number of highly correlated meta-analytic specifications may be examined simultaneously. Correcting only across specifications is not sufficient if the individual tests are poorly calibrated, while improving inference within each specification is not sufficient if researchers subsequently select significant results from the multiverse.

Permutation-based inference has already been proposed for several meta-analytic settings. Follmann and Proschan (Follmann & Proschan, 1999) developed permutation procedures for testing the average effect in random-effects meta-analysis; Higgins and Thompson (Higgins & Thompson, 2004) proposed permutation tests for moderator effects in meta-regression; Viechtbauer et al. (Viechtbauer et al., 2015) showed that permutation tests can provide adequate Type I error control when testing moderators in mixed-effects meta-regression; and Noma et al. (Noma et al., 2020) extended permutation inference to multivariate random-effects meta-analysis. These methods, however, were developed for inference within a single meta-analytic model rather than for inference across a multiverse of specifications.

Score-based sign flipping provides a natural route for connecting these two inferential layers. Instead of permuting outcomes or moderator values directly, sign-flipping methods operate on study- or observation-level score contributions. Under the null hypothesis, the ideal score contributions based on the true nuisance quantities have mean zero; replacing nuisance quantities by their estimates under the null yields the asymptotically equivalent score used in practice (De Santis et al., 2025a; Hemerik et al., 2020). Random sign flips of these contributions can then be used to construct a reference distribution for the test statistic, while subsequent developments have extended the same logic to simultaneous testing of many correlated models through permutation-based multiple-testing procedures (De Santis et al., 2025b).

In this paper, we extend score-based sign-flipping inference, and consequently the PIMA framework, to mixed-effects meta-analysis and meta-regression. We call this method

Post-Selection Inference in Multiverse Meta-Analysis (PIMMA). First, we derive study-level score contributions for testing meta-analytic regression coefficients while accounting for nuisance parameters and between-study heterogeneity. Second, we show how common sign-flipping transformations can be applied across multiple meta-analytic specifications to obtain global and specification-specific multiplicity-adjusted inference within the PIMA framework. The score-based sign-flipping inference for the meta-analysis model will be assessed using a simulation study compared with the common inferential methods. Finally, we illustrate the proposed approach by reanalyzing data from the multiverse meta-analysis of psychological treatments for depression proposed by Plessen et al. (2022) and subsequently conducted by Plessen et al. (2023), focusing on cognitive behavioral therapy for adults.

Mixed-Effects Meta-Regression

In this section, we present a general formulation of the mixed-effects meta-regression model, following the model formulation of Viechtbauer et al. (2015). In a meta-analysis with k independent studies, let y_i and v_i denote respectively the observed effect size estimate and the sampling variance for the i-th study. The model can be written as

$$\begin{aligned} \mathbf{y} &= \mathbf{D}\boldsymbol{\beta} + \mathbf{u} + \mathbf{e}, \\ \mathbf{u} &\sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I}), \\ \mathbf{e} &\sim \mathcal{N}(\mathbf{0}, \mathbf{V}), \end{aligned} \tag{1}$$

where $\mathbf{y}$ is the $k \times 1$ vector of observed effect size estimates, $\mathbf{D}$ is the $k \times p$ model matrix (including the intercept), and $\boldsymbol{\beta}$ is the vector of p regression coefficients. The vector $\mathbf{u}$ contains the study-level random effects and is assumed to follow a normal distribution with mean zero and variance τ^2, whereas $\mathbf{e}$ contains the within-study sampling errors. The sampling errors are assumed to be normally distributed with a diagonal variance-covariance matrix $\mathbf{V} = \text{diag}(v_1, \ldots, v_k)$, where the sampling variances v_i are treated as known. It follows that the marginal distribution of the observed effect size estimates is

$$\mathbf{y} \sim \mathcal{N}(\mathbf{D}\boldsymbol{\beta}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma} = \mathbf{V} + \tau^2\mathbf{I}. \tag{2}$$

For a fixed value of τ^2, the regression coefficients can be estimated by weighted least squares. Let $\mathbf{W} = \boldsymbol{\Sigma}^{-1} = (\mathbf{V} + \tau^2\mathbf{I})^{-1}$. The estimator of $\boldsymbol{\beta}$ is then

$$\mathbf{b} = (\mathbf{D}^\top\mathbf{W}\mathbf{D})^{-1}\mathbf{D}^\top\mathbf{W}\mathbf{y}. \tag{3}$$

Since τ^2 is unknown, the weights used in practice are obtained by replacing τ^2 with an estimate $\hat{\tau}^2$, so that $\hat{\mathbf{W}} = (\mathbf{V} + \hat{\tau}^2\mathbf{I})^{-1}$. Several estimators have been proposed for τ^2 (see Viechtbauer, 2005). We focus here on the DerSimonian–Laird estimator (1986), as a commonly used non-iterative method-of-moments estimator, and on restricted maximum likelihood (REML) method.

Both estimators can be expressed using the matrix

$$\mathbf{P} = \mathbf{W} - \mathbf{W}\mathbf{D}(\mathbf{D}^\top\mathbf{W}\mathbf{D})^{-1}\mathbf{D}^\top\mathbf{W}. \tag{4}$$

For the DerSimonian–Laird estimator, $\mathbf{W}$ is computed using only the inverse sampling variances, that is, $w_i = v_i^{-1}$. Since p denotes the total number of regression coefficients, including the intercept, the estimator is given by

$$\hat{\tau}^2_{\mathrm{DL}} = \frac{\mathbf{y}^\top\mathbf{P}\mathbf{y} - (k-p)}{\mathrm{tr}(\mathbf{P})}, \tag{5}$$

where negative $\hat{\tau}^2_{\mathrm{DL}}$ values are truncated to zero.

Likelihood-based estimators are obtained from the marginal normal model in (2). The log-likelihood of the random-effects model can be written as

$$\ell_{\mathrm{ML}}(\boldsymbol{\beta}, \tau^2) = -\frac{1}{2}\left[k\log(2\pi) + \log|\boldsymbol{\Sigma}| + (\mathbf{y} - \mathbf{D}\boldsymbol{\beta})^\top\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{D}\boldsymbol{\beta})\right]. \tag{6}$$

For a fixed value of τ^2, maximizing (6) with respect to $\boldsymbol{\beta}$ gives the weighted least squares estimator in (3). Substituting this estimator into the likelihood yields the profiled log-likelihood

$$\ell^{p}_{\mathrm{ML}}(\tau^2) = -\frac{1}{2}\left[k\log(2\pi) + \log|\mathbf{\Sigma}| + \mathbf{y}^\top\mathbf{P}\mathbf{y}\right]. \tag{7}$$

However, maximum likelihood estimators of variance components are generally downward biased because they do not account for the loss of degrees of freedom associated with estimating the fixed effects. Restricted maximum likelihood addresses this issue by maximizing a likelihood based on linear combinations of the data that remove the fixed effects. Up to an additive constant independent of τ^2, the restricted log-likelihood can be written as

$$\ell_{\mathrm{REML}}(\tau^2) \propto -\frac{1}{2}\left[\log|\mathbf{\Sigma}| + \log|\mathbf{D}^\top\mathbf{W}\mathbf{D}| + \mathbf{y}^\top\mathbf{P}\mathbf{y}\right]. \tag{8}$$

After $\hat{\tau}^2_{\mathrm{REML}}$ has been obtained, the final estimates of the meta-regression coefficients are computed using $\hat{\tau}^2_{\mathrm{REML}}$ in (3).

Several methods have been proposed to perform statistical inference on the regression coefficients in mixed-effects meta-regression models (see Viechtbauer et al., 2015). Here, we consider the three main approaches that will be compared in the simulation study: Wald-type inference, the Knapp–Hartung method (Knapp & Hartung, 2003), and permutations (Follmann & Proschan, 1999; Higgins & Thompson, 2004).

Under the standard Wald-type approach, the variance-covariance matrix of the estimated regression coefficients is estimated as

$$\hat{\mathbf{V}}_{\mathbf{b}} = (\mathbf{D}^\top\hat{\mathbf{W}}\mathbf{D})^{-1}, \tag{9}$$

where

$$\hat{\mathbf{W}} = (\mathbf{V} + \hat{\tau}^2\mathbf{I})^{-1}.$$

The standard error of b_j is given by the square root of the corresponding diagonal element of $\hat{\mathbf{V}}_{\mathbf{b}}$. The Wald-type test statistic is then defined as

$$z_j = \frac{b_j}{\sqrt{\widehat{\mathrm{Var}}(b_j)}}. \tag{10}$$

Under the null hypothesis $H_0 : \beta_j = 0$, z_j is compared against the critical value calculated at the α level of a standard normal distribution.

The Knapp–Hartung method uses the same coefficient estimates but replaces the standard variance-covariance matrix (9) with an adjusted version as $\hat{\mathbf{V}}_{\mathbf{b}\mathrm{KH}} = s^2\hat{\mathbf{V}}_{\mathbf{b}}$ with s^2 calculated as

$$s^2 = \frac{\mathbf{y}^\top\hat{\mathbf{P}}\mathbf{y}}{k-p}. \tag{11}$$

The test statistic is computed as in (10), using the corresponding adjusted standard error, and is compared against a t distribution with $k - p$ degrees of freedom. A truncated version of the Knapp–Hartung method sets $s^2 = 1$ whenever $s^2 < 1$, thereby preventing the adjusted standard error from being smaller than the standard Wald-type standard error.

Permutation testing provides a nonparametric alternative for testing moderator effects. Following the implementation in `metafor::permutest()` and the permutation approach discussed by Viechtbauer et al. (2015), for models with moderators the rows of the complete model matrix $\mathbf{D}$ are permuted while the observed effect-size estimates and sampling variances are kept fixed. For each permutation, the complete mixed-effects meta-regression model is refitted, including re-estimation of the regression coefficients and τ^2. For a given coefficient b_j, let $z_j^{(1)}$ denote the Wald-type statistic obtained from the original data and let $z_j^{(r)}$, $r = 2, \ldots, B$, denote the statistics obtained from $B - 1$ random permutations. The original data configuration is included as the first configuration. For a two-sided test, the permutation p value is computed as

$$p_{\mathrm{perm}} = \frac{\sum_{r=1}^{B} \mathbb{I}\left(|z_j^{(r)}| \geq |z_j^{(1)}|\right)}{B}. \tag{12}$$

When all unique permutations of the model matrix are enumerated, this gives the exact permutation distribution. Otherwise, the distribution is approximated using a finite number of random permutations.

Meta-regression using flipscores

The mixed-effects meta-regression model can be reformulated as a score-based test. This provides the link between moderator testing in meta-analysis and the sign-flipping score framework used by PIMA. The key change is that resampling is no longer applied to the rows of the model matrix, as in the permutation test described above, but to the study-level contributions to the score.

We focus on a single focal regression coefficient. We partition the complete design matrix as $\mathbf{D} = [\mathbf{X}, \mathbf{Z}]$, where $\mathbf{X}$ is the $k \times 1$ design vector for the focal predictor and $\mathbf{Z}$ contains the nuisance predictors, including the intercept when appropriate. The corresponding model can be written as

$$\mathbf{y} \sim \mathcal{N}\left(\mathbf{X}\beta + \mathbf{Z}\boldsymbol{\gamma}, \boldsymbol{\Sigma}\right), \qquad \boldsymbol{\Sigma} = \mathbf{V} + \tau^2\mathbf{I}. \tag{13}$$

We test $H_0 : \beta = \beta_0$, usually with $\beta_0 = 0$, while $\boldsymbol{\gamma}$ and τ^2 remain nuisance parameters. Under the null model, we estimate $\hat{\boldsymbol{\gamma}}_0$ and $\hat{\tau}_0^2$ and define

$$\widehat{\boldsymbol{\Sigma}}_0 = \mathbf{V} + \hat{\tau}_0^2\mathbf{I}, \qquad \widehat{\mathbf{W}}_0 = \widehat{\boldsymbol{\Sigma}}_0^{-1}. \tag{14}$$

The fitted values and residuals under the null are

$$\hat{\boldsymbol{\mu}}_0 = \mathbf{X}\beta_0 + \mathbf{Z}\hat{\boldsymbol{\gamma}}_0, \qquad \hat{\mathbf{r}}_0 = \mathbf{y} - \hat{\boldsymbol{\mu}}_0. \tag{15}$$

Although τ^2 is a nuisance parameter, it does not require an additional projection in the effective score. In the Gaussian meta-regression model, the parameters of the mean and the variance parameter τ^2 are Fisher-orthogonal,

$$\mathcal{I}_{\beta,\tau^2} = 0, \qquad \mathcal{I}_{\gamma,\tau^2} = \mathbf{0}.$$

This follows because the score for the mean parameters is linear in the residuals, whereas the score for τ^2 is quadratic in the residuals, and the corresponding cross-information is zero under the Gaussian model. Therefore, nuisance adjustment in the effective score is required for $\boldsymbol{\gamma}$, while τ^2 enters the test through the null working weights $\widehat{\mathbf{W}}_0$.

The connection with the general sign-flipping score framework follows directly from the Gaussian identity-link form of the meta-regression model. In the general framework, let $\mathbf{V}_Y = \mathrm{Var}(\mathbf{y})$ denote the response variance matrix, to distinguish it from the within-study sampling-variance matrix $\mathbf{V}$ used in meta-analysis. The working weight matrix is

$$\mathbf{W} = \mathbf{D}_\mu \mathbf{V}_Y^{-1} \mathbf{D}_\mu,$$

where $\mathbf{D}_\mu = \mathrm{diag}(\partial\mu_i/\partial\eta_i)$. For the Gaussian identity-link meta-regression model, $\mathbf{D}_\mu = \mathbf{I}$ and

$$\mathbf{V}_Y = \boldsymbol{\Sigma} = \mathbf{V} + \tau^2\mathbf{I}.$$

Therefore,

$$\mathbf{W} = \boldsymbol{\Sigma}^{-1}, \qquad \mathbf{V}_Y^{-1/2} = \mathbf{W}^{1/2}.$$

The general effective score of Hemerik et al. (2020) and De Santis et al. (2025a) therefore reduces directly to the following score for mixed-effects meta-regression. Define the weighted hat matrix for the nuisance design as

$$\mathbf{H}_Z = \widehat{\mathbf{W}}_0^{1/2}\mathbf{Z}\left(\mathbf{Z}^\top\widehat{\mathbf{W}}_0\mathbf{Z}\right)^{-1}\mathbf{Z}^\top\widehat{\mathbf{W}}_0^{1/2}. \tag{16}$$

The effective score, evaluated under H_0, is then

$$S_{\mathrm{eff}} = k^{-1/2}\mathbf{X}^\top\widehat{\mathbf{W}}_0^{1/2}(\mathbf{I}-\mathbf{H}_Z)\widehat{\mathbf{W}}_0^{1/2}\hat{\mathbf{r}}_0. \tag{17}$$

This is the direct specialization of the general flipscores effective score to the marginal Gaussian meta-regression model. Equivalently, it can be written as a sum of study-level effective score contributions,

$$S_{\text{eff}} = k^{-1/2} \sum_{i=1}^{k} \nu_i, \tag{18}$$

where ν_i is the contribution of study i to the effective score. Define the focal predictor residualized with respect to the nuisance predictors as

$$\widetilde{\mathbf{X}} = \mathbf{X} - \mathbf{Z}\left(\mathbf{Z}^\top \widehat{\mathbf{W}}_0 \mathbf{Z}\right)^{-1} \mathbf{Z}^\top \widehat{\mathbf{W}}_0 \mathbf{X}.$$

The contribution of study i can then be written as

$$\nu_i = \tilde{x}_i \hat{w}_{0i} \hat{r}_{0i}. \tag{19}$$

With a single focal moderator and an intercept-only nuisance model, this reduces to

$$\nu_i = (x_i - \bar{x}_w)\hat{w}_{0i}\hat{r}_{0i}, \qquad \bar{x}_w = \frac{\sum_{i=1}^{k} \hat{w}_{0i} x_i}{\sum_{i=1}^{k} \hat{w}_{0i}}. \tag{20}$$

Thus, in this case, the effective score is a weighted association between the centered moderator and the residuals under the null model. Under H_0, the ideal effective score contributions based on the true nuisance quantities have mean zero. Replacing the nuisance quantities by their estimates under the null yields the asymptotically equivalent score used in practice (De Santis et al., 2025a; Hemerik et al., 2020).

Let $\mathbf{F}^{(r)} = \operatorname{diag}(g_1^{(r)}, \ldots, g_k^{(r)})$ be a diagonal sign-flipping matrix, with $g_i^{(r)} \in \{-1, +1\}$ and $r = 1, \ldots, B$. The first flip is fixed as the identity, $\mathbf{F}^{(1)} = \mathbf{I}$, and the remaining $B - 1$ matrices are sampled at random. For flip r, the flipped effective score is

$$S_{\text{eff}}^{(r)} = k^{-1/2} \mathbf{X}^\top \widehat{\mathbf{W}}_0^{1/2} (\mathbf{I} - \mathbf{H}_Z) \widehat{\mathbf{W}}_0^{1/2} \mathbf{F}^{(r)} \hat{\mathbf{r}}_0, \tag{21}$$

or, equivalently,

$$S_{\text{eff}}^{(r)} = k^{-1/2} \sum_{i=1}^{k} g_i^{(r)} v_i. \tag{22}$$

Following the standardized sign-flip score test, each flipped score is standardized using the leading term of its flip-specific conditional variance (De Santis et al., 2025a; Girardi et al., 2024). In the present meta-regression setting, this term is estimated as

$$\widehat{C}^{(r)} = k^{-1}\mathbf{X}^{\top}\widehat{\mathbf{W}}_0^{1/2}(\mathbf{I}-\mathbf{H}_Z)\mathbf{F}^{(r)}(\mathbf{I}-\mathbf{H}_Z)\mathbf{F}^{(r)}(\mathbf{I}-\mathbf{H}_Z)\widehat{\mathbf{W}}_0^{1/2}\mathbf{X}. \tag{23}$$

The standardized two-sided statistic is therefore

$$T^{(r)} = \left| \frac{S_{\text{eff}}^{(r)}}{\sqrt{\widehat{C}^{(r)}}} \right|. \tag{24}$$

The sign-flipping p value for a single specification is

$$p_{\text{flip}} = \frac{\sum_{r=1}^{B} \mathbb{I}\left(T^{(r)} \geq T^{(1)}\right)}{B}. \tag{25}$$

The asymptotic validity of this meta-regression specialization is inherited from the general standardized sign-flipping score framework under the corresponding regularity conditions. As customary in conventional mixed-effects meta-analysis, the within-study sampling variances v_i are treated as known. We further assume that the meta-regression mean model is correctly specified, that the complete design matrix has full rank and fixed dimension as k increases, that the estimated working weights converge to finite and non-degenerate limits, and that the score contributions satisfy the regularity and Lindeberg conditions of De Santis et al. (2025a), excluding asymptotically dominating studies.

In particular, for the heterogeneity estimator used under the null, assume that

$$\hat{\tau}_0^2 \xrightarrow{P} \tilde{\tau}^2$$

for some finite $\widetilde{\tau}^2 \geq 0$. For each study, continuity then gives

$$\hat{w}_{0i} = \frac{1}{v_i + \hat{\tau}_0^2} \xrightarrow{P} \frac{1}{v_i + \widetilde{\tau}^2} = \widetilde{w}_i.$$

Thus, estimation of τ^2 produces a convergent working-weight sequence of the type allowed by the general sign-flipping framework, provided the remaining regularity conditions hold. Importantly, $\widetilde{\tau}^2$ need not correspond to the true variance structure. De Santis et al. (2025a) show that, provided that the mean model is correctly specified and the required regularity conditions hold, the standardized sign-flip score test remains asymptotically valid under general variance misspecification.

Under these conditions, the difference between the flip-specific leading variance term used for standardization and the corresponding conditional variance vanishes in probability as $k \to \infty$. Therefore, the standardized study-level sign-flipping score test for $H_0 : \beta = \beta_0$ is asymptotically an α-level test.

The proposed test therefore inherits the robustness of the flipscores framework to misspecification of the study-level variance structure, provided that the meta-regression mean model and the other regularity conditions are correctly specified. This robustness does not extend to misspecification of the mean model or to violations of the assumed independence structure.

This test keeps the null model fixed and constructs the reference distribution by sign-flipping the study-level effective score contributions. For each tested null hypothesis, the constrained null meta-regression model is fitted once, and the B sign-flipping transformations are then applied directly to the resulting score contributions without re-estimating the regression coefficients or τ^2. This differs from `metafor::permutest()`, where the complete meta-regression model, including τ^2, is refitted for every permutation (Higgins & Thompson, 2004).

We explored this computational difference in the supplementary materials. For a meta-analysis with $k = 100$ studies and 3 predictors, `metafor::permutest()` takes roughly 14 seconds, compared with 0.3 seconds for the score sign-flipping approach. The score sign-flipping test is implemented in the `flipmeta` R package (https://github.com/filippogambarota/flipmeta). From now, we call `flipmeta` the extension of the flipscores framework to the meta-analysis model.

From `flipmeta` to the multiverse meta-analysis

The score sign-flipping test provides inference for a single meta-analytic specification. PIMA extends this logic to a multiverse by applying the same sign-flipping transformations across all admissible specifications and combining the resulting standardized score statistics across specifications (Girardi et al., 2024). We consider a fixed finite set of S meta-analytic specifications, indexed by $s = 1, \ldots, S$, while the asymptotic results refer to an increasing number of independent studies.

Specifications may differ in inclusion criteria, effect-size metric, moderator coding, heterogeneity estimator, or other analytical choices, provided that each specification can be represented within the independent mixed-effects meta-regression framework described above and tests a substantively compatible focal null hypothesis. Let β_s denote the focal coefficient in specification s and define

$$H_s : \beta_s = 0.$$

The global null hypothesis is the intersection of the specification-specific null hypotheses,

$$H_{\text{global}} = \bigcap_{s=1}^{S} H_s = \{\beta_s = 0 \text{ for all } s = 1, \ldots, S\}. \tag{26}$$

Rejecting this hypothesis implies that the focal effect is non-null in at least one specification, but does not identify which specification is responsible for the rejection (Girardi et al., 2024).

To define common resampling transformations across specifications, let

$$\mathcal{I} = \{1, \ldots, K\}$$

denote the master set of independent studies occurring in the multiverse, and let $\mathcal{I}_s \subseteq \mathcal{I}$ denote the subset of studies included in specification s. For each transformation $r = 1, \ldots, B$, a single sign $g_i^{(r)} \in \{-1, +1\}$ is generated for each study $i \in \mathcal{I}$, with $g_i^{(1)} = 1$ for all i. The sign-flipping matrix used in specification s is then

$$\mathbf{F}_s^{(r)} = \operatorname{diag}\left\{g_i^{(r)} : i \in \mathcal{I}_s\right\}. \tag{27}$$

Hence, whenever a study occurs in multiple specifications, the same sign is applied to its score contribution in all of them. If a specification excludes a study, the corresponding element of the common sign vector is simply omitted. This is the construction used by PIMA for specifications involving different subsets of observations (Girardi et al., 2024). The common sign-flipping transformations preserve the joint resampling dependence among specification-level statistics induced by their shared study-level information.

For each specification s, its constrained null model is fitted and the standardized statistic $T_s^{(r)}$ is computed for each common transformation $r = 1, \ldots, B$. Inference across the multiverse can then be performed at two levels. First, a global test evaluates H_{global}. Following PIMA, the flipped statistics can in principle be combined using different non-decreasing functions; for post-selection inference, a particularly useful choice is the max-T statistic (Girardi et al., 2024),

$$M^{(r)} = \max_{s=1,\ldots,S} T_s^{(r)}. \tag{28}$$

The corresponding global p value is

$$p_{\text{global}} = \frac{\sum_{r=1}^{B} \mathbb{I}\left(M^{(r)} \geq M^{(1)}\right)}{B}. \tag{29}$$

Rejecting the global null implies that the focal parameter is non-null in at least one specification, providing weak control of the family-wise error rate (Girardi et al., 2024).

The validity of this construction follows directly from PIMA once the specification-specific `flipmeta` tests satisfy the conditions described in the previous section. For a fixed finite number of specifications, if each specification satisfies the conditions required for the asymptotic validity of its standardized score test, the number of independent studies in each specification increases, and the same study-level signs are used across specifications, the resulting PIMMA global test is asymptotically an α-level test (Girardi et al., 2024). Specifications containing different subsets of studies are accommodated by restricting the common sign vector to the studies retained in each specification.

Second, selective inference can be performed for individual specifications. PIMA obtains multiplicity-adjusted p values through the max-T framework, which can be implemented using single-step or sequential step-down variants (Westfall & Troendle, 2008; Westfall & Young, 1992). Under the joint resampling results established for PIMA, these procedures provide strong control of the family-wise error rate, allowing inference on particular specifications selected after inspecting the multiverse (Girardi et al., 2024). This strong control follows from the joint resampling structure and the max-T procedure, rather than merely from accounting for dependence among the specification-level statistics.

Simulation study

We performed a simulation study to evaluate the inferential performance of the proposed `flipmeta` approach and compare it with standard Wald-type inference and the Knapp–Hartung method (Knapp & Hartung, 2003). Standard permutation-based procedures for meta-analysis (Follmann & Proschan, 1999; Higgins & Thompson, 2004) were not included because they require repeated permutation of the model matrix and refitting of the complete meta-analytic model, resulting in a prohibitive computational cost.

Data were generated following the general simulation framework of Viechtbauer et al. (2015). We considered meta-analyses consisting of k independent studies, each comparing an experimental group (E) with a control group (C) on a quantitative outcome. For study $i = 1, \ldots, k$, observations in the two groups were assumed to follow normal distributions with means μ_i^E and μ_i^C and a common within-study standard deviation σ_i. The true standardized mean difference for study i was therefore

$$\theta_i = \frac{\mu_i^E - \mu_i^C}{\sigma_i}.$$

For each simulated study, we computed the observed standardized mean difference as

$$d_i = \frac{\bar{x}_i^E - \bar{x}_i^C}{s_i},$$

where $\bar{x}_i^E$ and $\bar{x}_i^C$ are the observed group means and s_i is the pooled standard deviation. The approximately unbiased estimator of θ_i was then obtained using the small-sample correction (Hedges, 1981),

$$y_i = \left(1 - \frac{3}{4\left(n_i^E + n_i^C\right) - 9}\right) d_i. \tag{30}$$

The corresponding sampling variance was estimated as

$$v_i = \frac{1}{n_i^E} + \frac{1}{n_i^C} + \frac{y_i^2}{2\left(n_i^E + n_i^C\right)}.$$

The vector of true study effects was generated according to the mixed-effects meta-regression model

$$\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \mathbf{u},$$

where $\mathbf{X}$ contains the focal predictors, $\boldsymbol{\beta}$ their corresponding regression coefficients, $\mathbf{Z}$ contains nuisance predictors, $\boldsymbol{\gamma}$ their coefficients, and $\mathbf{u}$ is the vector of study-specific random effects. The random effects were independently generated as $u_i \sim \mathcal{N}(0, \tau^2)$.

Under the null hypothesis, the focal coefficients were set to $\boldsymbol{\beta} = \mathbf{0}$, allowing empirical Type I error rates to be evaluated. Under the alternative hypothesis, focal regression coefficients were set to $\beta \in \{0.1, 0.3, 0.5\}$, representing increasingly strong associations and allowing statistical power to be evaluated. The nuisance coefficients were similarly varied over $\gamma \in \{0.1, 0.3, 0.5\}$ under both the null and alternative hypotheses.

Focal and nuisance predictors were jointly generated from a multivariate normal distribution with mean zero and unit marginal variances. A compound-symmetry correlation structure with pairwise correlation $\rho = 0.5$ was used, so that focal and nuisance predictors were moderately correlated. This setting was chosen to represent the common situation in which inference on a focal moderator must account for correlated study-level covariates.

Residual heterogeneity was varied by setting $\tau^2 \in \{0, 0.01, 0.10, 0.50\}$. The condition $\tau^2 = 0$ corresponds to the absence of residual between-study heterogeneity, whereas the remaining values represent progressively larger amounts of heterogeneity on the standardized mean-difference scale. The number of studies was varied over $k \in \{10, 20, 40, 100\}$, covering meta-analyses ranging from relatively small to large. Study sample sizes were also allowed to vary across studies. For each study, equal expected group sizes were assumed, such that n_i^E and n_i^C were generated from the same negative-binomial distribution. Four average group sizes were considered, $n \in \{10, 30, 50, 80\}$, and the negative-binomial dispersion parameter was chosen so that the variance of the generated sample sizes was approximately twice their mean. This produces heterogeneous study sizes while retaining a prespecified average sample size, more closely reflecting the variation in study precision typically encountered in meta-analysis.

For each combination of simulation parameters, the resulting effect-size estimates y_i and sampling variances v_i were analyzed using the competing inferential procedures. Performance under the null hypothesis was quantified by the empirical Type I error rate at the nominal significance level $\alpha = 0.05$, whereas performance under the alternative hypothesis was quantified by empirical statistical power.

The Figure 1 illustrate the results

To summarise, the `flipmeta` method has optimal control of the type-1 error in every condition in addition to a good statistical power. As reported by other simulation studies (Viechtbauer et al., 2015) the type-1 error is not controlled by the Wald test in case of model misspecification (equal-effects model when $\tau^2 \neq 0$) or with a low number of studies ($k = 10$ in our simulation). At least in this specific simulation, for a single meta-analyis there is no a real advantage of using `flipmeta`. However, when moving to the multiverse case, the `flipmeta` can easily provide weak and strong control of the FWER without assuming a dependency structure among the tests. Without `flipmeta` one could use standard methods such as Bonferroni or Holm. These methods control the FWER under arbitrary dependence, but do not exploit the dependence structure among tests. This can result in an important reduction in statistical power when tests are correlated.

In multivariate tests, and in particular in a multiverse analysis, the tests are likely to be correlated. Consider a multiverse analysis with two specifications based on the same set of k studies, differing only in whether an equal-effects model (i.e., fixing $\tau^2 = 0$) or a random-effects model is fitted. The score vectors, and consequently the sign-flipped test statistics, are likely to be strongly correlated, especially when τ^2 is close to zero. Considering (19), for an intercept-only random-effects model under $H_0 : \mu = 0$, there are no nuisance predictors, $\tilde{x}_i = 1$, and $\hat{r}_{0i} = y_i$. The score contribution of the i-th study therefore simplifies to $v_i = \hat{w}_{0i} y_i$. Since y_i is identical across the two specifications, differences in the score contributions arise only from differences in the study weights, which in this example are mainly determined by the different values of τ^2. As $\hat{\tau}^2$ approaches zero, the random-effects weights approach the equal-effects weights, making the corresponding score vectors increasingly similar.

As a practical example, let's use the dataset from the meta-analysis by Credé et al. (2010) about the relationship of class attendance with grades and student characteristics. The dataset is included in the `metafor` R package (Viechtbauer, 2010).

The Code 1 shows that correlation in this specific multiverse meta-analysis is relatively high, especially when only changing the type of model. Applying a BonHolmferroni correction here would be too conservative reducing the statistical power. The actual correlation could be close to zero but is very likely that in real-world multiverse meta-analysis the scenarios are not independent. In the next section we will se the empirical comparison between the Holm correction and the maxT correction in a real multiverse with hundreds of scenarios.

Application to psychotherapy for depression

Dataset and original investigation

We illustrate PIMMA using the database assembled for the multiverse meta-analysis of psychological treatments for depression. The study was first described in a preregistered protocol by Plessen et al. (2022) and was subsequently completed and reported by Plessen et al. (2023). Its purpose was to determine whether conclusions about the efficacy of psychotherapy for depression were robust across the many defensible decisions available to meta-analysts. The authors searched PubMed, Embase, PsycINFO, and the Cochrane Central Register of Controlled Trials for randomized controlled trials published up to January 1, 2022. Studies comparing a psychological treatment with a control condition were eligible without restrictions on the type of psychotherapy, target population, treatment format, control condition, or definition of depression. The resulting database comprised 1,206 effect-size estimates from 415 studies and 71,454 participants. In the original investigation, combinations of inclusion criteria and statistical methods produced 4,281 meta-analyses. The results generally supported the efficacy of psychotherapy, while also showing that effect estimates varied systematically with choices such as the control condition, risk-of-bias restrictions, and adjustment for publication bias (Plessen et al., 2023).

The present application is not intended to reproduce that full multiverse. Instead, it uses a clinically coherent subset of the database to demonstrate post-selection inference across closely related meta-analytic specifications. We restricted the analysis to studies of cognitive behavioral therapy in adult populations. Before accounting for dependence among effects from the same study, this subset contained 275 effect-size estimates from 86 studies.

Data preparation

The available standardized mean differences were expressed as Hedges' g, and their sampling variances were obtained from the corresponding standard errors. Records without an available effect-size estimate were excluded. Study identifiers and the categories used to define the multiverse—including treatment and control conditions, treatment format, target population, outcome rater, and risk of bias—were harmonized before applying the eligibility criteria. Risk of bias was represented by three categories: low risk, some concerns, and high risk.

Because a study could contribute multiple effects, the eligible effects were first selected separately within each specification and then aggregated to obtain one effect-size estimate per study. Aggregation incorporated an assumed correlation among effects from the same study. This ensured that the study, rather than the individual outcome or measurement occasion, was the independent unit contributing to each meta-analysis, in line with the study-level score contributions required by PIMMA.

Multiverse specifications

All specifications estimated an intercept-only random-effects model, with the pooled Hedges' g as the parameter of interest. The multiverse varied four decisions governing the evidence included in each meta-analysis and two decisions concerning within- and between-study variance (Table 1). Crossing these decisions yielded $3 \times 3 \times 4 \times 3 \times 3 \times 2 = 648$ candidate specifications. We retained specifications containing at least six independent studies, leaving 438 eligible specifications based on 73 distinct study subsets and between 6 and 86 studies per specification.

Note. The application was restricted to cognitive behavioral therapy in adult populations. Unrestricted levels retained all categories available in the selected data, including other control conditions, high-risk-of-bias studies, and other treatment formats.

Results

In our multiverse with 438 scenarios, roughly 96% of the raw p-values were significant. After correction using the maxT approach, 86% of all scenarios remained significant. This corresponds to 90% of the initially significant results surviving the correction, or a loss of 10% of the initially significant results. The main advantage of this approach becomes clearer when compared with the Holm method, which is the most powerful procedure controlling the FWER without making assumptions about the dependency structure among the tests. In fact, none of the scenarios remained significant after applying the Holm correction.

Discussion

Conclusion

Limitations and future perspectives

The present work has some limitations mainly related to the extension of the `flipscores` method to the meta-analysis model. The main problem is about assuming that the meta-analytic dataset has k independent studies contributing with a single effect size (i.e., a two-level model). In practice, most of the time there is dependency among observed effects due to multilevel and/or multivariate data structure (Cheung, 2013; Van den Noortgate et al., 2013). Some recent works extended the flipscores approach to mixed-effects models with nested and crossed random-effects (Andreella et al., 2025; Vesely & Andreella, 2026) developing an appropriate permutation approach considering the dependency of observations. We are working on the implementation of this approach also for complex meta-analitic data structure.

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Code 1 Correlations between test statistics in a multiverse meta-analysis

```
library(metafor)
devtools::load_all()
set.seed(24617)

#library(flipmeta)

dat <- dat.crede2010
# calculating Fisher z transformed correlation coefficient
dat <- escalc("ZCOR", ri = ri, ni = ni, data = dat)
dat$id <- 1:nrow(dat)

# scenario 1, full data, random-effects
fit1 <- rma(yi, vi, data = dat, method = "REML")

# scenario 2, only journals articles
# (excluding dissertations and other sources), random-effects
fit2 <- rma(yi, vi, data = dat[dat$source == "journal", ])

# scenario 3, full data, equal-effects (tau2 = 0)
fit3 <- rma(yi, vi, data = dat, method = "EE")

fitl <- list(fit1, fit2, fit3)

# multiverse analysis, 1000 permutations
res <- flipmeta(fitl, B = 1000, id = "id", progress = FALSE)

# extracting the B x 3 matrix of permuted t statistics
# and calculating the correlations

R <- cor(res$Tspace)
round(R, 2)

# raw p values
res$summary_table$p
# maxT (step down) p values
p.adjust(res)$summary_table$p.adj
# bonferroni p values
stats::p.adjust(res$summary_table$p, method = "holm")
##              mod1:intrcpt mod2:intrcpt mod3:intrcpt
## mod1:intrcpt         1.00         0.83         0.61
## mod2:intrcpt         0.83         1.00         0.53
## mod3:intrcpt         0.61         0.53         1.00
## [1] 0.001 0.001 0.001
## [1] 0.001 0.001 0.001
## [1] 0.003 0.003 0.003
```

Table 1
Specification decisions defining the application multiverse.

Specification decision	Levels
Control condition	Waitlist; care as usual; unrestricted
Risk of bias	Low risk only; low risk or some concerns; unrestricted
Treatment format	Individual; group; guided self-help; unrestricted
Outcome rater	Clinician-rated; self-reported; unrestricted
Within-study correlation	ρ = .30, .50, or .70
Heterogeneity estimator	Restricted maximum likelihood; DerSimonian–Laird

Figure 1

Simulation results

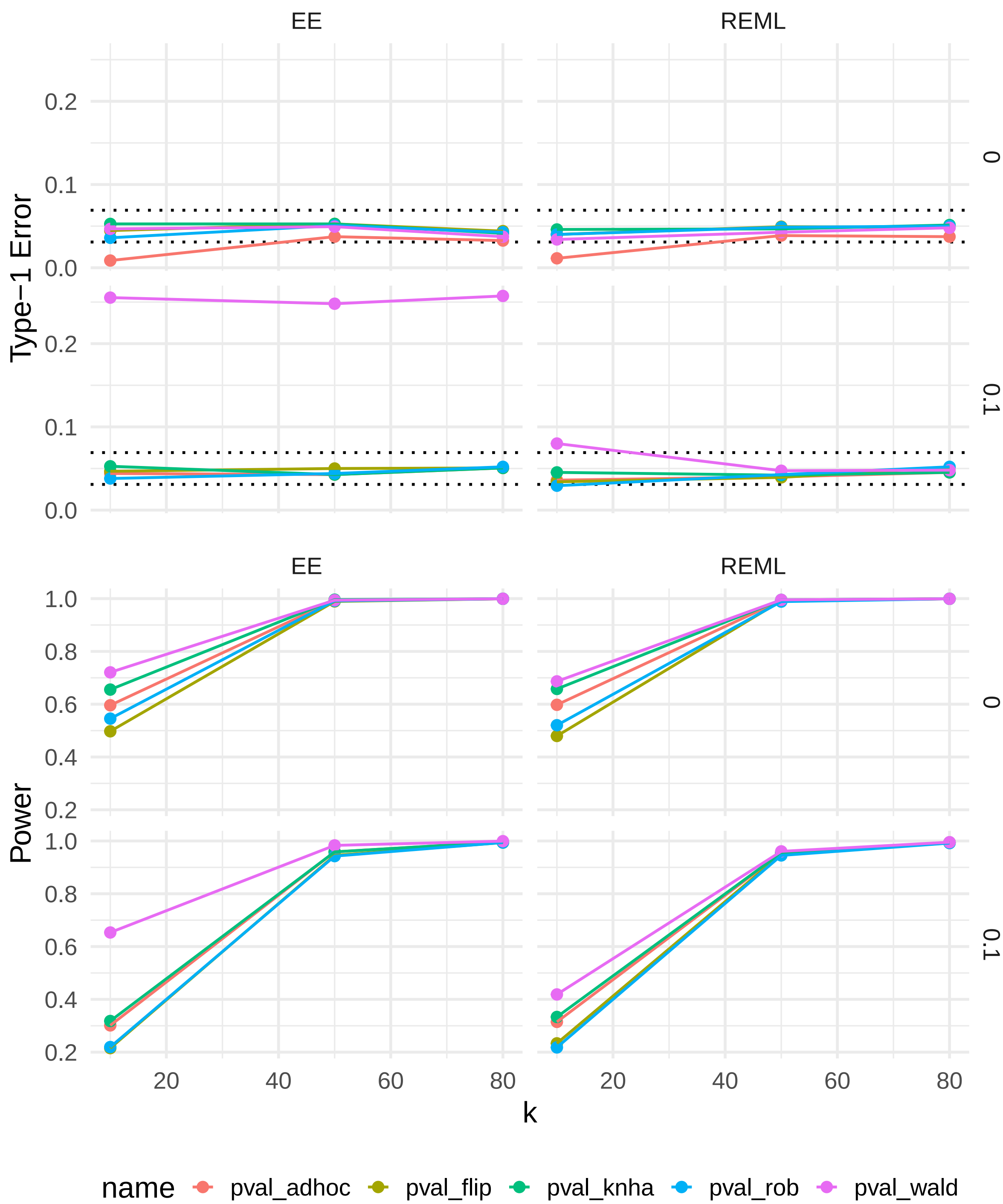
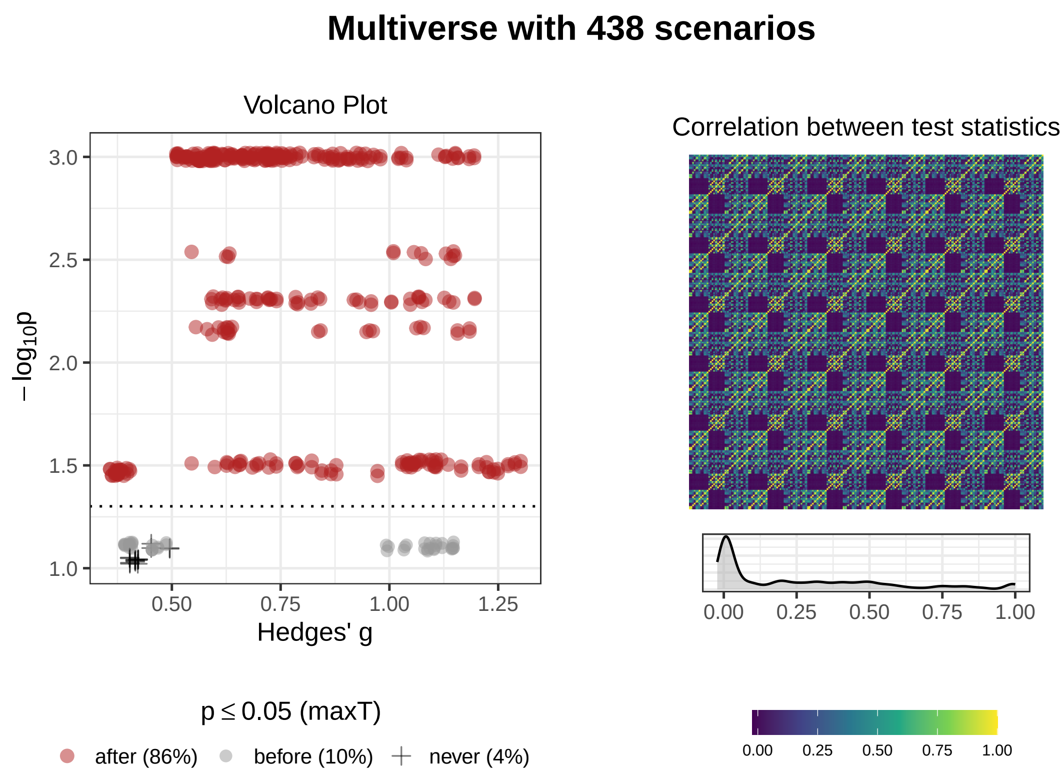


Figure 2


Multiverse with 438 scenarios
Volcano Plot
3.0
2.5
2.0
1.5
1.0
$-\log_{10}p$
0.50
0.75
1.00
1.25
Hedges' g
Correlation between test statistics
0.00
0.25
0.50
0.75
1.00
$p \leq 0.05$ (maxT)
after (86%)
before (10%)
never (4%)
0.00
0.25
0.50
0.75
1.00